

CUSTOMER INTELLIGENCE

Segmentation, anomaly exploration, dimensionality reduction, and calibrated default-risk modeling

ML PORTFOLIO CASE STUDY

github.com/Awais-Asghar/customer-intelligence-ml

Four questions, four methods

01

PCA

What structure explains most variation?

ROLE

Compression and visualization

02

K-Means

Which broad behaviors repeat?

ROLE

Descriptive segmentation

03

DBSCAN

Which records are locally sparse?

ROLE

Anomaly exploration

04

Boosting

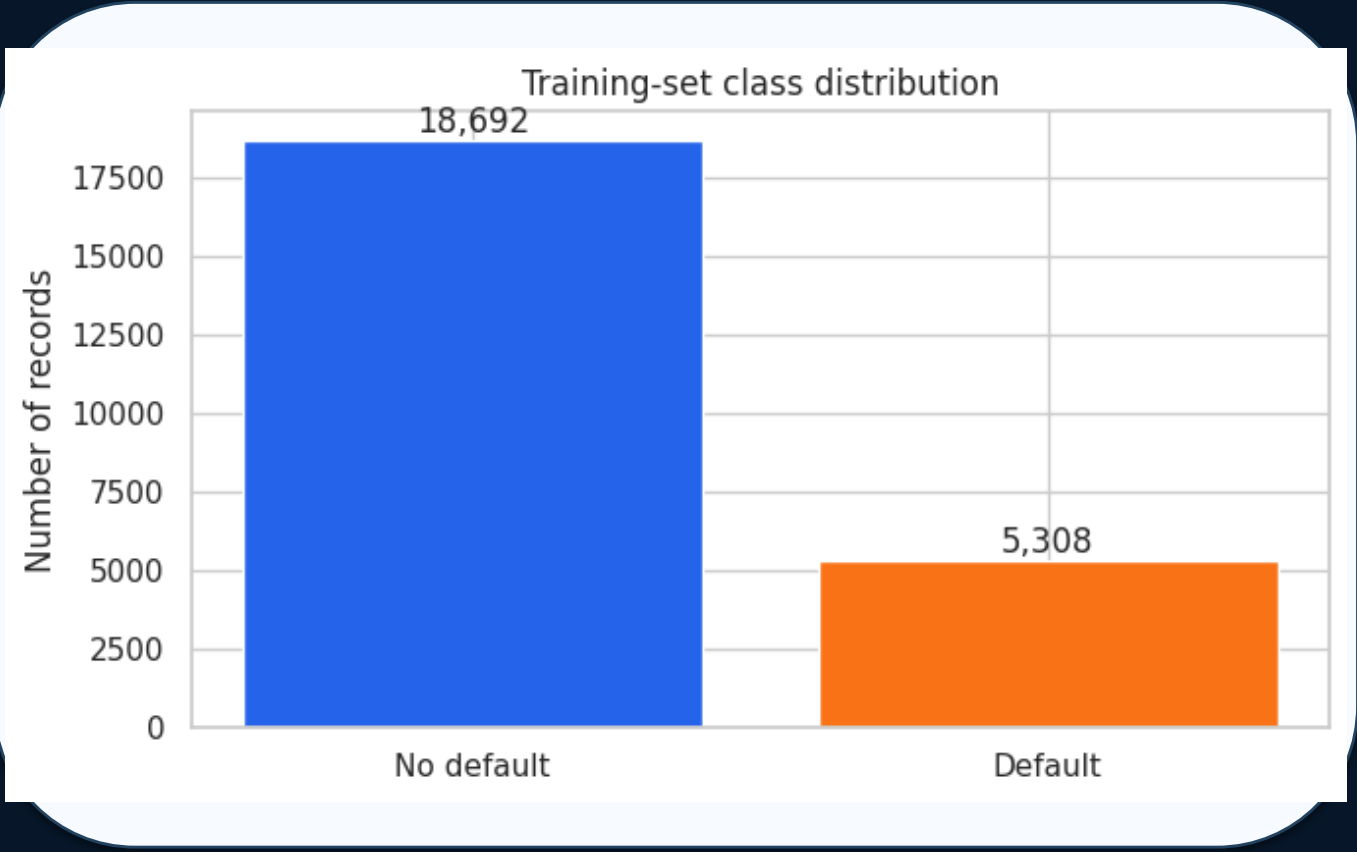
How uncertain is historical default?

ROLE

Calibrated risk estimation

Clusters are not risk classes. Density noise is not fraud. Predictions are probabilities, not decisions.

A realistic, imbalanced credit dataset



30,000
CUSTOMER RECORDS

23
RAW EXPLANATORY FEATURES

22.12%
HISTORICAL DEFAULTS

0
MISSING VALUES

WHY ACCURACY IS NOT ENOUGH

A model that predicts only the majority class would appear accurate while missing every historical default.

Source UCI Default of Credit Card Clients, dataset 350, CC BY 4.0

Target prevalence 6,636 positive records

The test set stays closed until the end

TRAIN

24,000 records | 80%

Candidate selection
Three-fold stratified CV

VALIDATION

3,000 records | 10%

Threshold selection
No candidate tuning

TEST

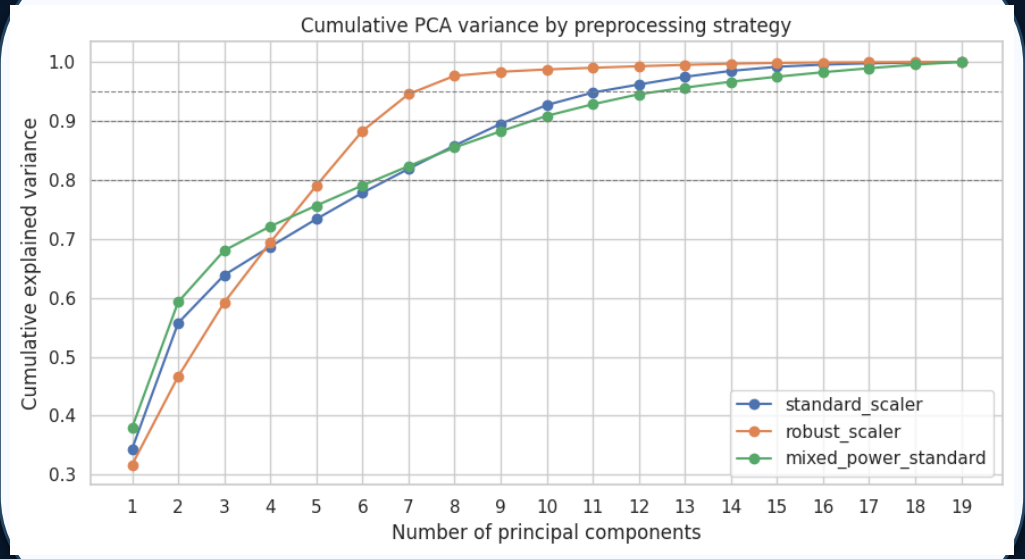
3,000 records | 10%

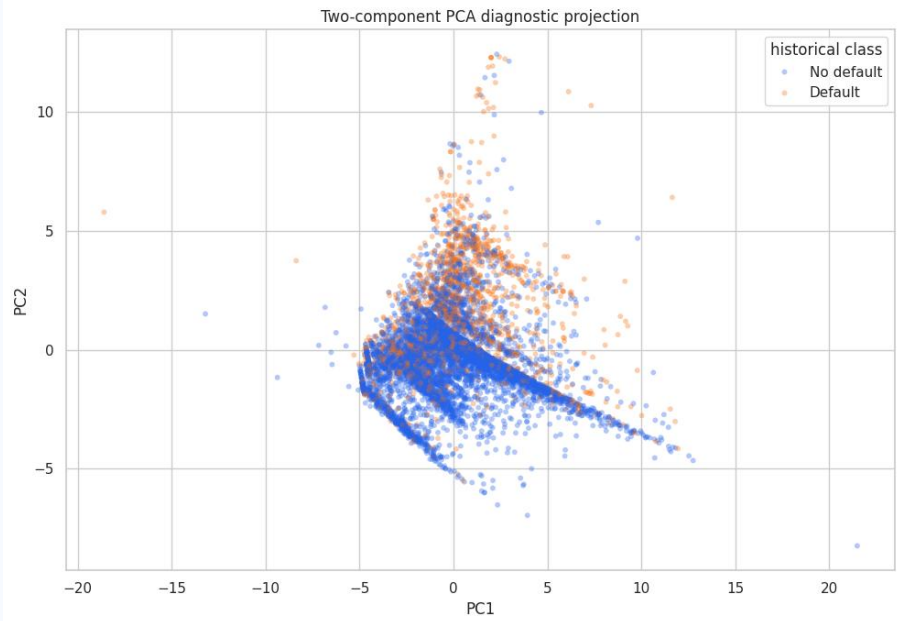
Opened once
Final evaluation only

AUDIT RESULTS

0	IDENTIFIER OVERLAP
0.0742	MAXIMUM ABSOLUTE SMD
10	TEST ROWS MATCHING A DEV PROFILE
21	PROFILES WITH CONFLICTING LABELS

PCA retains signal without inventing certainty





19
BEHAVIORAL INPUTS

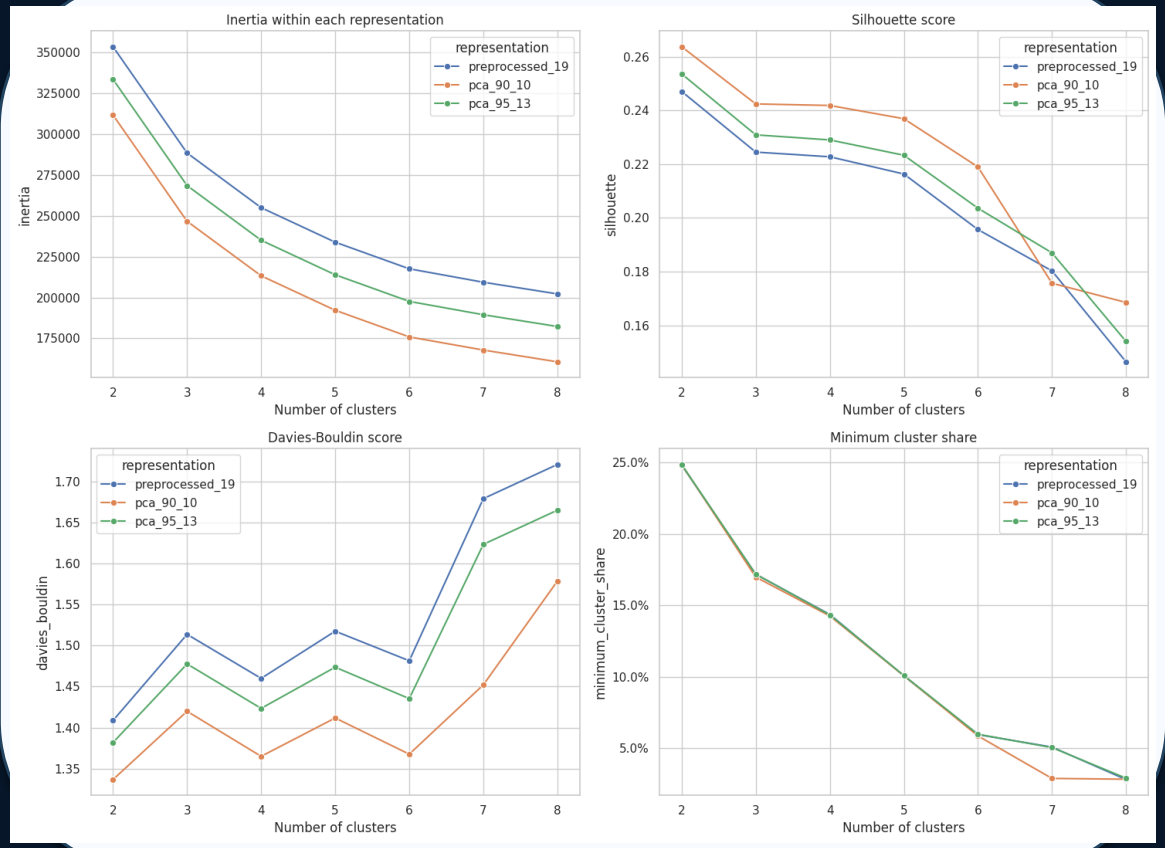
13
COMPONENTS RETAINED

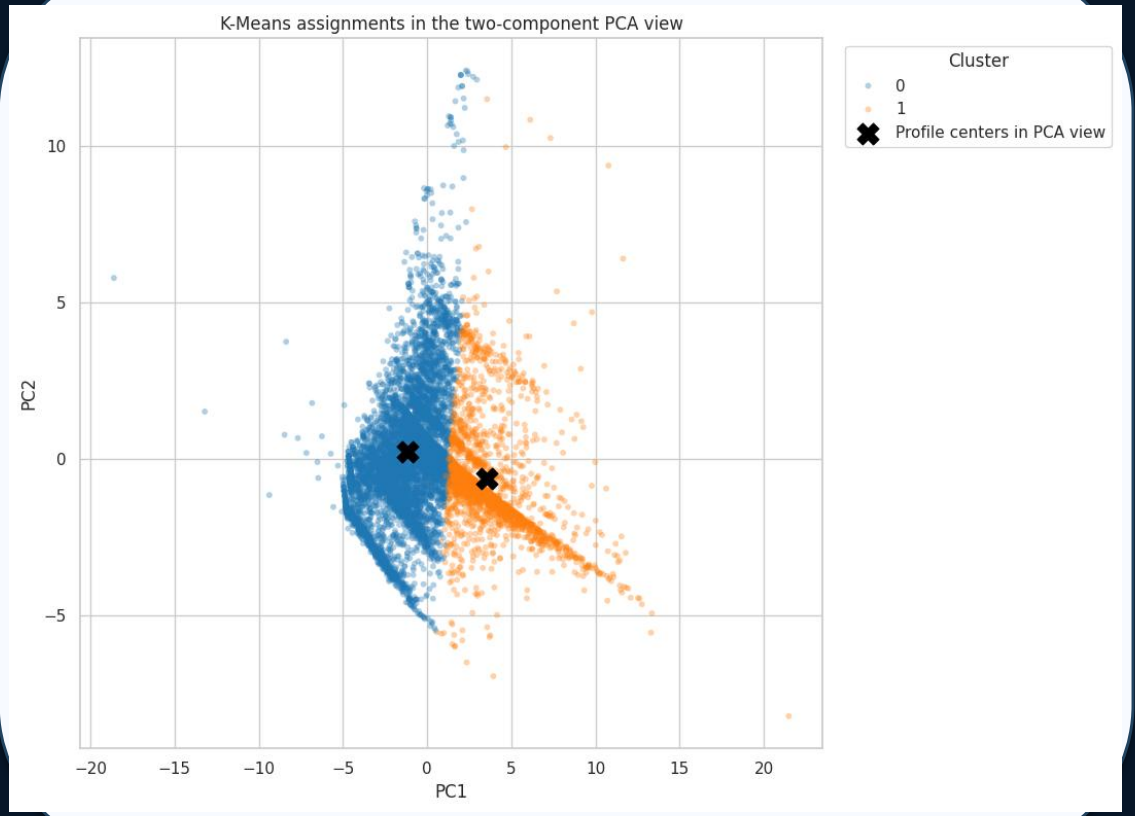
95.63%
CUMULATIVE VARIANCE

59.29%
FIRST TWO COMPONENTS

The 13-component representation supports PCA and clustering analysis. The supervised model uses 54 engineered behavioral features.

Two stable groups, with meaningful overlap





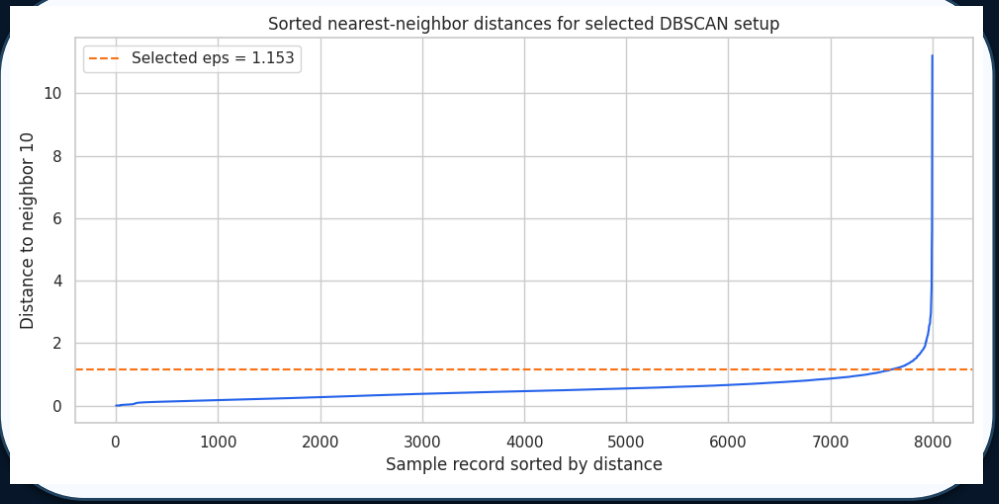
2
SELECTED CLUSTERS

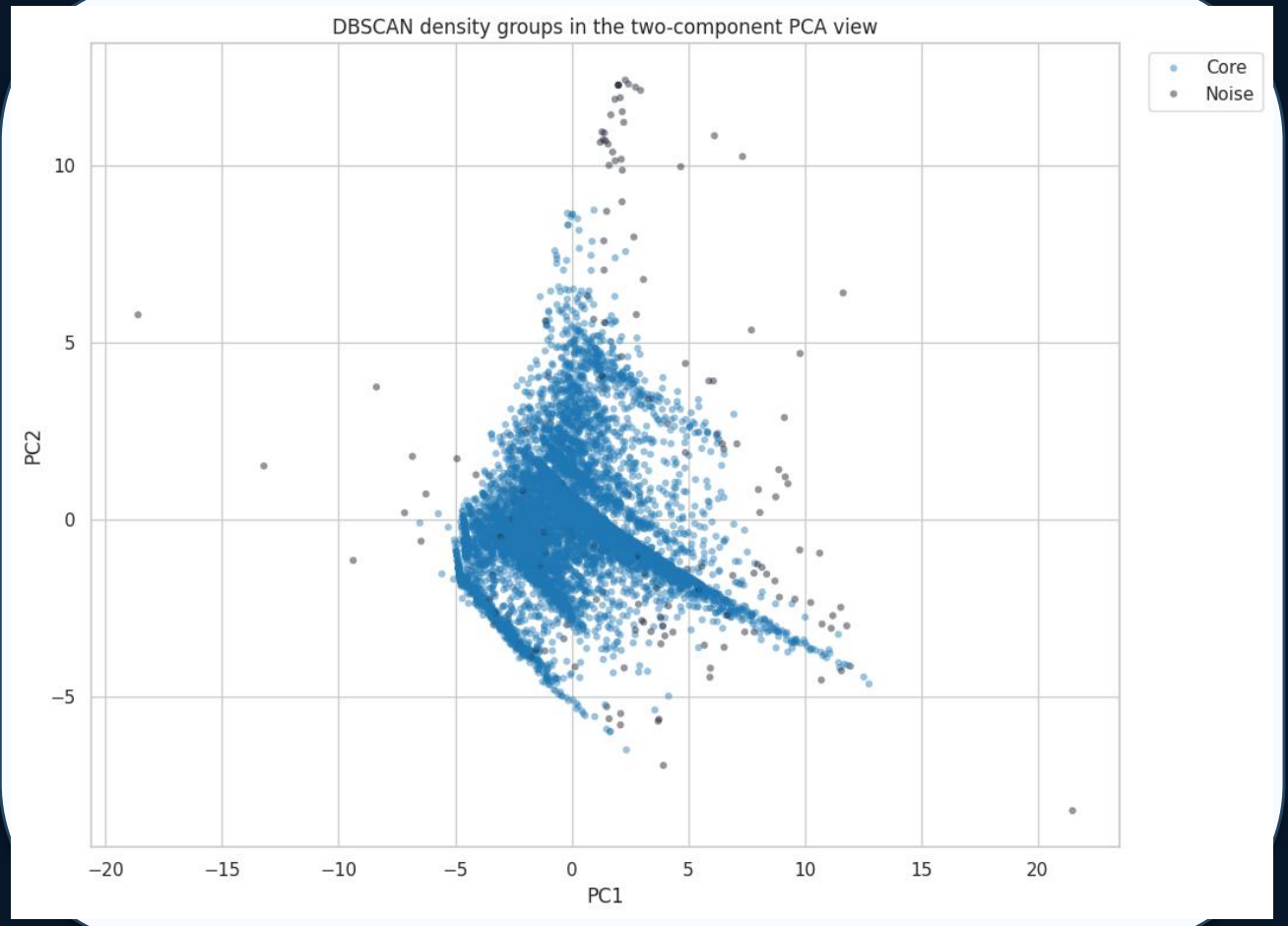
0.2638
SILHOUETTE SCORE

0.9960
STABILITY ARI

24.66%
SMALLEST CLUSTER

DBSCAN reveals sparse records, not new segments





SELECTED CONFIGURATION

- 5 whitened PCA components
- eps 1.1526 and min_samples 10
- One substantial core group
- 2.08% density-noise records

Interpretation: unusual multivariate behavior, not fraud, wrongdoing, or a guaranteed future outcome.

A supervised pipeline designed for honest comparison

01

Behavioral inputs

19 non-demographic variables

02

Feature engineering

54 deterministic features

03

Candidate search

10 HGB configurations

04

Selection

Three-fold stratified CV

MODEL SELECTION OBJECTIVE

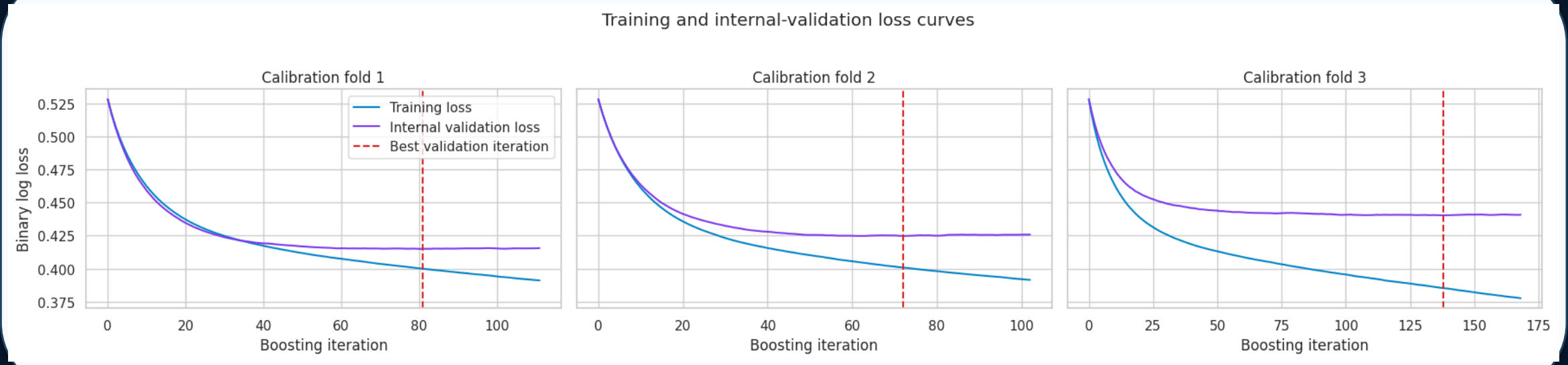
PR-AUC

Selected because the positive class is 22.12% of the data

Demographics excluded from prediction

0.05	15	600	30
LEARNING RATE	MAX LEAF NODES	MAX ITERATIONS	EARLY-STOP PATIENCE

Early stopping controls unnecessary boosting



81
FOLD 1 BEST ITERATION

72
FOLD 2 BEST ITERATION

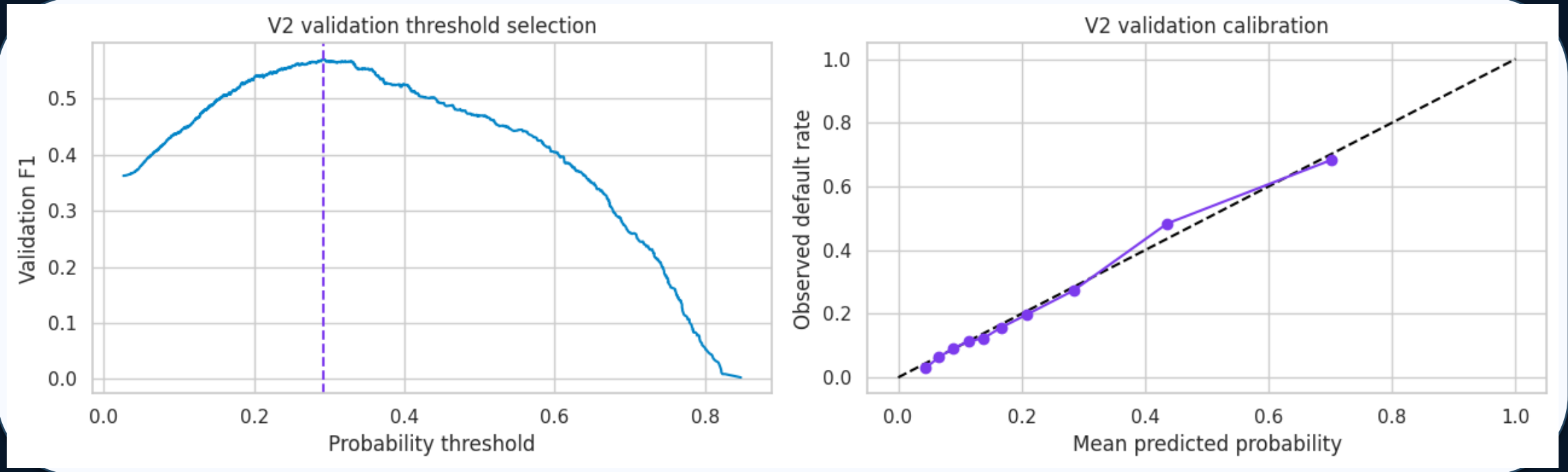
138
FOLD 3 BEST ITERATION

600
MAXIMUM CAP

Final calibrated estimators — **116, 131, and 106 iterations**

No fold reached the cap

Threshold selection is an operating decision



0.2916

VALIDATION-SELECTED THRESHOLD

GAINS

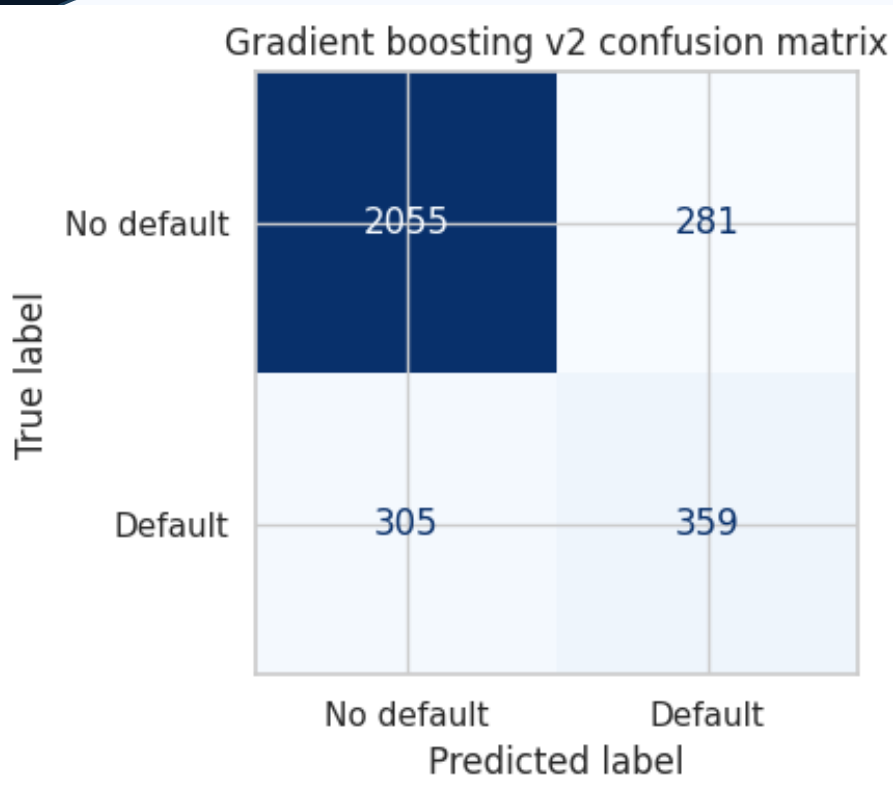
Recall, balanced accuracy, and F1

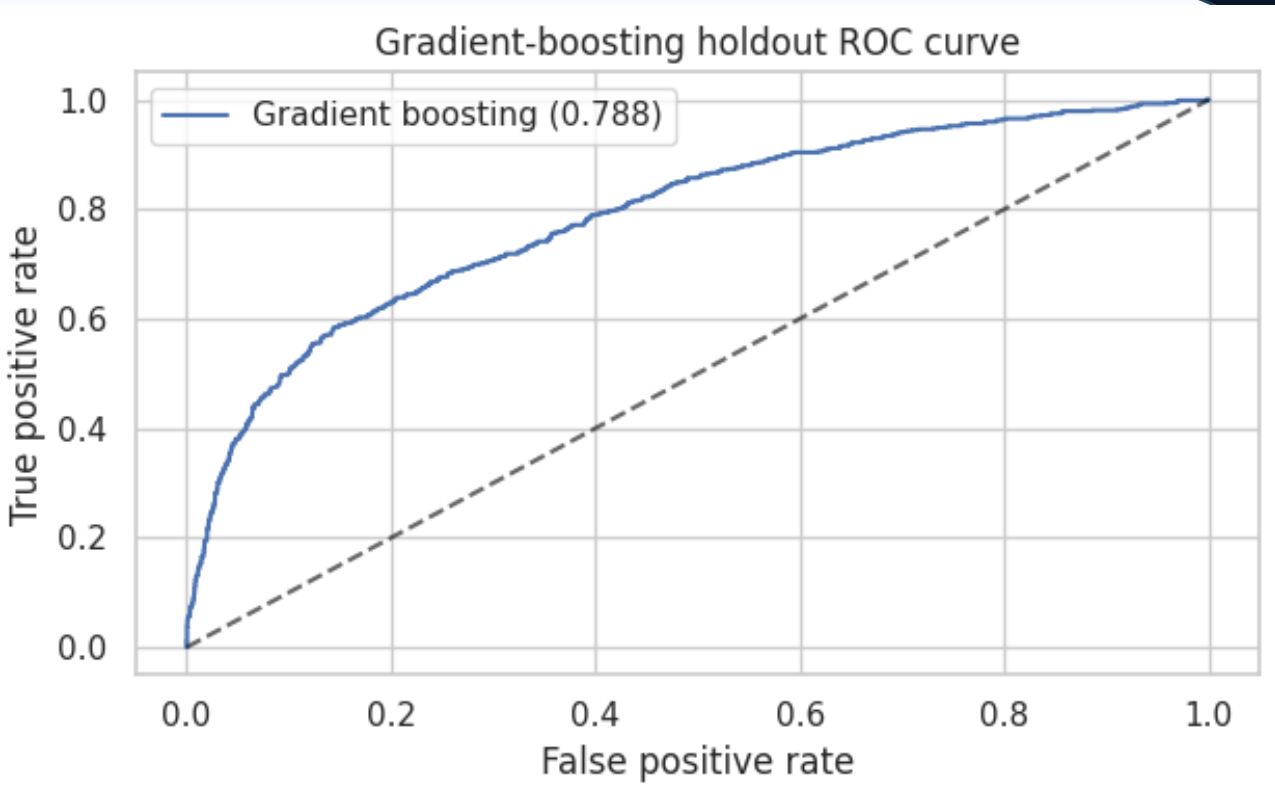
TRADEOFF

Lower precision and overall accuracy

ROC-AUC and PR-AUC do not change when only the decision threshold changes.

The final test result is useful and honest





80.47%	71.02%	56.09%	54.07%	55.06%
ACCURACY	BALANCED ACCURACY	PRECISION	RECALL	F1 SCORE

What the project proves, and what it does not

LESSONS

- Prevent leakage before optimizing metrics
- Use separate methods for separate questions
- Tune the threshold on validation data only
- Treat negative results as useful evidence
- Communicate uncertainty and operating tradeoffs

LIMITS

- Conflicting labels prevent deterministic prediction
- Rare categories increase uncertainty
- Clusters are descriptive, not causal
- DBSCAN noise is not evidence of fraud
- Production use requires governance and monitoring

REPOSITORY **github.com/Awais-Asghar/customer-intelligence-ml**

Notebook | Tests | Figures | Reproducibility